

interconnected architecture like the brain. Its building block is not a logic gate—like in a digital computer—but a silicon neuron whose behaviour and connectivity are programmable. Read more about the programmable silicon neuron at <http://www.stanford.edu/group/brainsinsilicon/about.html#Emulate>.

Bio-replication. Some also use the term 'bio-replication' interchangeably with 'bio-morphic.' However, there is believed to be a subtle distinction. Bio-morphic mimics the underlying structure, whereas bio-replication completely reproduces it. The only difference between the natural and artificial designs in the case of replication is brought about by the difference in material.

Biomimetics. As a middle path, we have biomimetics, which involves mimicking a natural function with certain features of the natural structure that make the function possible reproduced in the design.

"Biomimetics utilises natural techniques for inspiration, and not the absolute blueprints. The needs of any individual organism are likely to be vastly different from our own, and most adaptations are reached through compromises. A given eye design may have good anti-reflection properties, but they may have been tuned specifically for a small cone of vision, or be restricted in the wavelengths of operation," comments Dr Judith Braganca, BITS Pilani, KK Birla Campus, Goa.

"Biomimetics provides an immense database of designs that can inspire creative thought. This technique evolves and grows with the development in research and technology much like the evolution of nature," he adds.

Dr Braganca works with a group of salt-loving micro-organisms that are known to inhabit salt pans. These grow under stress conditions of high solar radiation, 25 per cent salt and sometimes anoxic (anaerobic) conditions. They are brightly-pigmented with red, orange and pink pigments.

"These pigments have been implicated to protect the organisms from

sunlight as their production can be increased with exposure to light. So, we are in the process of designing a sunscreen with these haloarchaeal pigments for human use so it will protect the skin from ultraviolet (UV) radiation," she explains.

Why let nature solve our problems?

The strongest reasons cited in favour of nature-inspired solutions are that they are likely to be evolutionary and sustainable. "Biology is really the only technology that we know is sustainable. The more we understand and exploit biological solutions, the more sustainable our technology will become. In the neuro realm, this means lower power, higher robustness, and the ability to scale to the nano-scale, despite increasing heterogeneity and randomness," says Dr Boehen.

Jayakrishnan opines that solutions based on nature are also very robust and adaptive. He says, "With over two billion years of research and development, nature has evolved highly-efficient materials, structures, tools, mechanisms, processes, algorithms, methods and systems. Design approaches inspired by nature often result in systems or processes that exhibit much greater robustness in performance in unstructured environments compared to existing ones."

Dr Lakhtakia points out that there might be certain advantages in hindsight too. "Imitation of a natural solution could be beneficial as it may open up a possibility not thought of earlier. For instance, one could copy the shape of a certain natural structure for optical purposes but later find that the shape confers mechanical robustness as well under certain conditions. In pharmaceutical ethnobotany, a certain drug isolated from a certain plant used traditionally for the treatment of a particular ailment may turn out to be useful against another ailment as well. Thus, there may be an additional advantage, but only in hindsight," he explains.

Respect and learn, do not underestimate

While it sounds all hunky-dory, nature-inspired theories, designs and products are not as easy as the simplified accounts of Newton's or Archimedes' discoveries that we read in primary school. It involves systematic research, a multi-disciplinary approach and lots of risk too.

In the course of the discussion, Jayakrishnan makes it clear that to gain inspiration and guidance from nature is not an easy task. It involves multidisciplinary research and a certain degree of reverse engineering too.

"The creations of nature are biological models subjected to principles of physics, chemistry, mechanical engineering, materials science and many other branches of science and engineering. So, it is required to perform a highly systematic reverse engineering by dividing the system or process based on each category of the principles involved, mathematically modelling and identifying an analogy for each from the artificial world," he says.

Dr Braganca agrees, "Creating any nature-inspired design or product requires an amalgamation of many sciences. To replicate the photomechanical heat-sensing mechanism of the beetle *Melanophila acuminata* for creating a biomimetic infrared sensor, one would require expertise in both the biology and engineering," she says.

So, in general, mapping the concepts learnt from nature to the physical realm is a huge challenge. Nature has many complex creatures that have various ways of structure, survival and signalling. Applying these to product and design applications involves manifold challenges, which are not to be underestimated while undertaking a biomimetic project. Of course, it is obvious that the benefits make the efforts worthwhile. So, wish these researchers all the best! ●

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